

Demystifying the IoT for Embedded Design Engineers



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With so much of buzz around the Internet of Things (IoT), electronics engineers are keen to know what it really is all about. The primary purpose of this article is to explain implementation aspects of the IoT from a designer's perspective. It explains, at architecture level, aspects that designers should look into while designing.

Let us start by busting a few myths surrounding the IoT.

Myth #1. IoT devices are different and need special processors.

(This is not true; existing designs can be upgraded for IoT applications.)

Myth #2. All IoT devices need to have Internet Protocol version 6 (IPv6) protocol-compliant network software stack.

(You can have IoT devices that have proprietary communication ports without networking stack software but still can function as IoT devices.)

Myth #3. IoT devices need sophisticated software.

(The IoT is a solution framework and most applications typically run on the central server. Devices can be simple data collectors that send data across to the central servers.)

Let us now see some important aspects of the IoT, which should help electronics engineers in understanding it better.

Defining the IoT

The IoT is basically a global infrastructure enabling advanced services by con-

necting existing and evolving interoperable communication technologies. It is classified under two broad categories:

1. People-to-things (customer to business, or C2B)

2. Things-to-things, *aka*, machine-to-machine, or M2M (mostly business-to-business)

Each of these categories needs to be handled differently. People-to-things typically involves devices at home, fitness-focused applications, connected white goods and the whole gamut of wearables (such as arm bands). While M2M covers most of the automation segment, especially in the manufacturing industry. We need to understand that the IoT is not input- but output-driven mostly by applications.

The IoT architecture

An IoT system consists of three elements:

1. Sensors
2. Connectivity
3. People and processes

The IoT depends on sensors to provide data from the environment. Connectivity ensures that data is delivered to the central sever in which the applications run. Finally, applications pick up the data and process it to produce meaningful output.

Let us look into the IoT architecture in a little more detail. Fig. 1 shows how the IoT system is structured.

First element at the bottom is Things.

These could be devices with sen-

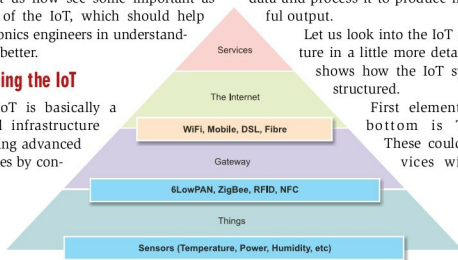


Fig. 1: Structure of the IoT system